

School of Computer Science and Artificial Intelligence

Lab Assignment # 5.2

Program	: B. Tech (CSE)
Specialization	: --
Course Title	: AI Assisted coding
Course Code	: 23CS002PC304
Semester	II
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Task 1:

- Zero-shot: Prompt AI with only the instruction. Write a Python function to determine whether a given number is prime

Prompt: Write a Python function to determine whether a given number is prime.

```
def is_prime(n):  
    if n <= 1:  
        return False  
  
    for i in range(2, int(n ** 0.5) + 1):  
        if n % i == 0:  
            return False  
  
    return True  
  
print(is_prime(2)) # True  
print(is_prime(3)) # True
```

```
C:\Users\atur\OneDrive\Desktop\Ai_assistanr>C:/Python314/pyt  
True  
True
```

Explanation:

- The function `is\_prime` checks if a number `n` is prime by first checking if it is less than or equal to 1, which are not prime numbers.
- Then, it iterates from 2 to the square root of `n`, checking if `n` is divisible by any of those numbers.
- If it finds a divisor, it returns False; otherwise, it returns True, indicating that `n` is prime.

Task 2:

- One-shot: Provide one example: Input: [1, 2, 3, 4], Output: 10 to help AI generate a function that calculates the sum of elements in a list.

Prompt: Write a Python function to calculate the sum of all elements in a list.

```
def calculate_sum(numbers):  
    total = 0  
    for num in numbers:  
        total += num  
    return total  
  
print(calculate_sum([1, 2, 3, 4]))
```

```
C:\Users\atur\OneDrive\Desktop\Ai_ass  
10
```

Explanation:

- The function initializes a variable total to 0 and iterates through each element in the list.
- It adds each number to total using a loop.
- Finally, it returns the total sum of all elements in the list

Task 3:

Few-shot: Give 2–3 examples to create a function that extracts digits from an alphanumeric string.

Prompt: Write a Python function that extracts only the digits from an alphanumeric string.

```
def extract_digits(text):  
    digits = ""  
    for char in text:  
        if char.isdigit():  
            digits += char  
    return digits  
  
print(extract_digits("abc123"))  
print(extract_digits("a1b2c3"))  
print(extract_digits("xyz7890test"))
```

```
C:\Users\atur\OneDrive\Desktop\Ai_assistanr>C:/Py  
123  
123  
7890
```

Explanation:

- The function iterates through each character in the string.
- It checks whether the character is a digit using `isdigit()`.
- If true, it appends the digit to a new string and finally returns only the extracted digits.

Task 4:

- Compare zero-shot vs few-shot prompting for generating a function that counts the number of vowels in a string.

Prompt: Write a Python function to count the number of vowels in a string.

```
def count_vowels(text):  
    vowels = "aeiouAEIOU"  
    count = 0  
    for char in text:  
        if char in vowels:  
            count += 1  
    return count  
  
print(count_vowels("hello"))  
print(count_vowels("AEIOU"))  
print(count_vowels("Python"))
```

```
C:\Users\atur\OneDrive\Desktop\Ai_assistanr>python  
2  
5
```

Explanation:

- Compare zero-shot vs few-shot prompting for generating a function that counts the number of vowels in a string

Task 5:

- Use few-shot prompting with 3 sample inputs to generate a function that determines the minimum of three numbers without using the built-in min() function.

Prompt:

Write a Python function to find the minimum of three numbers without using the built-in min() function.

```
def minimum_of_three(a, b, c):  
    minimum = a  
  
    if b < minimum:  
        minimum = b  
  
    if c < minimum:  
        minimum = c  
  
    return minimum  
  
print(minimum_of_three(3, 7, 5))  
print(minimum_of_three(10, 2, 8))  
print(minimum_of_three(4, 4, 9))
```

```
C:\Users\atur\OneDrive\Desktop\Ai_assistant>C:/Python3  
3  
2  
4
```

Explanation:

- The function assumes the first number is the minimum and stores it in a variable.
- It then compares the second and third numbers one by one and updates the minimum if a smaller value is found.
- Finally, it returns the smallest of the three numbers without using min().